

Emotion-Aware Autonomous Driving: A VR-Based Multimodal Analysis Using Gaze and Physiological Signals

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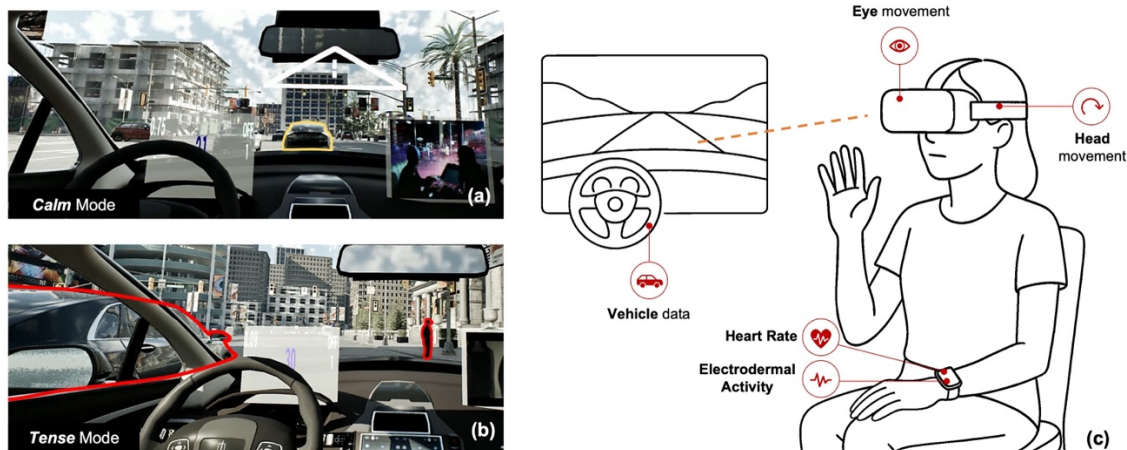


Figure 1: Screenshots from the VR-based autonomous driving simulation. (a) Calm Mode: The vehicle performs a smooth, gradual stop after detecting a lead vehicle's sudden halt, minimizing discomfort. (b) Tense Mode: The vehicle makes an abrupt evasive maneuver and stop, inducing urgency and discomfort. (c) A schematic overview of the experimental setup, showing multimodal data collected from the participant, including vehicle dynamics, eye and head movements, heart rate (BVP), and electrodermal activity (EDA).

Abstract

While autonomous driving technology continues to advance, understanding passengers' emotional experiences remains limited. This study examines emotion recognition in fully autonomous scenarios using a virtual reality (VR) simulation with two contrasting driving styles—Calm and Tense. Multimodal data analysis revealed significant behavioral and physiological differences between the

two modes. A Random Forest classifier achieved 97.1% accuracy in distinguishing passenger emotional states, demonstrating the feasibility of emotionaware autonomous driving systems.

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1 Introduction

Autonomous driving is advancing rapidly, but understanding passengers' emotional experiences remains limited. In fully autonomous scenarios (SAE Level 5), the absence of passenger control makes it difficult to infer internal states such as anxiety or discomfort from observable behaviors [Dillen et al., 2020]. As passengers

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shift from active participants to passive observers, attention-based emotion recognition shows clear limitations. Virtual reality (VR) has been proposed as a safe method to simulate high-risk driving [Lee et al., 2022], yet most VR-based studies emphasize presence or usability rather than modeling emotional discomfort in realistic traffic conditions [Doula et al., 2023]. This study aims to quantify and predict passengers' emotional states in autonomous driving scenarios by leveraging gaze behavior and physiological responses within a VR environment. To this end, we designed two driving conditions: a Calm mode, characterized by smooth deceleration, and a Tense mode, marked by abrupt and discomfort-inducing braking events (Figure 1). Based on the collected data, we conducted statistical analyses and developed a Random Forest-based classification model to distinguish between two driving modes. The results showed that multimodal features—especially gaze behavior, head dynamics, and physiological signals—can reliably predict passenger discomfort in fully autonomous 2 driving scenarios. The key contributions of our work are: (1) a VR simulation framework that systematically elicits passenger comfort and discomfort, (2) a multimodal dataset collected under emotionally salient conditions, and (3) a Random Forest-based affect recognition model that demonstrates the feasibility of emotion-aware autonomous driving.

2 Our approach

2.1 VR autonomous driving simulation

We developed a VR simulation environment in CARLA (Unreal Engine) and implemented it on the HTC Vive Focus 3 with an integrated eye tracker for precise gaze tracking. Two driving styles—Calm and Tense—each included four highrisk traffic scenarios such as sudden braking, abrupt cut-ins, and unexpected pedestrian appearances. Multimodal data were continuously recorded, including vehicle dynamics, gaze and head movements, surrounding vehicle positions, and physiological signals (EDA, BVP) via the Embrace-Plus device.

2.2 Experiments

Participants experienced Calm and Tense driving modes in immersive autonomous driving scenarios, with gaze, head movements, and biosignals recorded via a head-mounted display and sensors. Thirty-two licensed adults took part under consistent conditions with IRB approval.

2.3 Analysis of behavioral and physiological responses

Statistically significant differences were observed between the Calm and Tense conditions across all major indicators. The pupil diameter, averaged over a 10-second window following event onset, was larger in the Tense condition ($M = 4.077$ mm) than in the Calm condition ($M = 3.927$ mm), $t = 26.007$, $p < .001$. Participant P22 showed a maximum dilation of 0.527 mm in the Tense condition, accompanied by an increase in EDA; the coefficient of variation reached 9.4%, compared to 3.75% in the Calm condition. Vertical head movement (Y-axis displacement) was 63% greater in the Tense condition ($M = 25.688$) than in the Calm condition ($M = 15.725$), $t = 7.117$, $p < .001$. Participants also tended to shift gaze leftward,

lower their heads, and rotate more along the yaw axis. A Random Forest classifier trained on multimodal features (pupil diameter, gaze, and head movements) achieved 97.1% mean accuracy with 5-fold cross-validation. Feature importance identified head yaw rotation, horizontal head position, and gaze origin along the x-axis as the top predictors.

3 Conclusion and discussion

This study demonstrates the potential of emotion-aware autonomous driving by classifying passenger states from gaze and physiological signals in VR-based scenarios. Future work will apply deep learning models, including Transformers and Graph Neural Networks, to improve temporal modeling and prediction accuracy.

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